Vascular Sonography Cheat Sheet

Prepared for ARDMS Vascular Specialty Study

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1 Vascular Sonography Terms

Term	Definition
Acoustic Impedance	The resistance of a medium to the propagation of sound waves, calculated as density times propagation speed.
Acoustic Shadowing	An artifact where sound waves are blocked by a strong reflector, resulting in a dark area distal to the structure.
Acoustic Speckle	The granular appearance in ultrasound images due to interference of scattered echoes.
Aliasing	An artifact in Doppler ultrasound where high velocities appear in the opposite direction due to insufficient sampling rate (Nyquist limit).
Amplification	The process of increasing the strength of received echo signals in ultrasound systems.
Amplitude	The maximum displacement or height of a sound wave from its equilibrium position.
Aneurysm	Abnormal dilation of a blood vessel wall, often due to weakening; can be fusiform or saccular.
A-Mode	Amplitude mode; a display mode showing echo amplitude versus depth along a single line.
Anechoic	Echo-free area in an ultrasound image, appearing black, typically indicating fluid-filled structures.
Angle Correct	Adjustment in Doppler ultrasound to account for the angle between the sound beam and blood flow direction for accurate velocity measurement.
Antegrade	Flow proceeding in the normal forward direction (e.g., toward the periphery in arteries).
Aperture	The effective diameter of the transducer active area that transmits or receives sound waves.
Apodization	Technique to reduce side lobes by varying the excitation of array elements.
Array	A group of transducer elements arranged in a line or curve for beam forming.
Artifacts	False or misleading information in ultrasound images not representing actual anatomy.
Atherosclerosis	Buildup of plaque in arteries, leading to narrowing and hardening.
Attenuation	Loss of ultrasound intensity as it travels through tissue due to absorption, reflection, and scattering.
Attenuation Coefficient	Measure of how much ultrasound intensity decreases per unit distance in tissue (dB/cm/MHz).
Autocorrelation	Signal processing technique used in color Doppler to estimate mean velocity and variance.

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Term	Definition
Axial Resolution	Ability to distinguish two structures along the beam axis; improved by higher frequency.
Backscatter	Reflection of sound waves back to the transducer from small scatterers like RBCs.
Bandwidth	Range of frequencies in a pulse; wide bandwidth improves axial resolution.
BART	Blue Away, Red Toward; convention for color Doppler flow direction.
Baseline	Zero velocity line in spectral Doppler display; shifting it can help resolve aliasing.
Beam Former	Electronic circuitry that controls the timing of pulses to steer and focus the beam.
Bernoulli's Principle	Principle stating that an increase in the speed of a fluid occurs simultaneously with a decrease in pressure or a decrease in the fluid's potential energy, used to estimate pressure gradients in vascular flow.
Bi-directional Doppler	Doppler system capable of displaying flow toward and away from the transducer.
Blossoming	Artifact in color Doppler where color bleeds outside vessel walls due to high gain.
B-Mode	Brightness mode; 2D grayscale imaging where echo strength is displayed as brightness.
Boundary Layer	Thin layer of fluid adjacent to the vessel wall where flow velocity is reduced due to friction.
Bruit	Audible sound caused by turbulent blood flow, often due to stenosis.
Calf Muscle Pump	Mechanism where calf muscles compress veins to aid venous return during walking.
Cavitation	Formation and collapse of gas bubbles in tissue due to high ultrasound intensity.
Clutter	Low-frequency noise in Doppler signals from tissue motion.
Coded Excitation	Technique using coded pulses to improve signal-to-noise ratio and penetration.
Color Doppler Imaging (CDI)	Overlay of color on B-mode image representing flow velocity and direction.
Color Priority	Setting that determines when color is displayed over grayscale echoes.
Comet Tail	Artifact appearing as a series of linear echoes distal to a strong reflector like metal.
Compensation (TGC)	Time gain compensation; amplifies deeper echoes to compensate for attenuation.
Compression	Reduction of dynamic range to fit signals into display range.

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Term	Definition
Continuity Equation	Principle that flow volume is constant; used to calculate area from velocity (Q = A * V).
Continuous Wave (CW) Doppler	Doppler mode using constant transmission; no range resolution but high velocity detection.
Contrast Harmonics	Imaging using harmonic frequencies generated by contrast agents.
Contrast Resolution	Ability to distinguish differences in echo intensity.
Critical Stenosis	Degree of narrowing where flow becomes significantly reduced, often >70-80Cross Talk
Leakage of signals between channels in spectral Doppler, causing mirror image artifact.	
Crystal	Piezoelectric element in transducer that converts electrical to acoustic energy.
Curie Point	Temperature at which piezoelectric material loses its properties.
Curved Sequence Array	Transducer array with curved elements for sector imaging.
Damping Material	Material behind crystal to shorten pulse duration, improving axial resolution.
Dead Zone	Area near transducer where imaging is poor due to ringing.
Decibels (dB)	Logarithmic unit measuring intensity ratios in ultrasound.
Demodulation	Process to extract Doppler shift from received signals.
Density	Mass per unit volume; affects acoustic impedance.
Depth Ambiguity	Artifact in CW Doppler where depth of flow is unknown.
Destructive Interference	When waves out of phase cancel each other.
Diastolic Reversal	Backward flow in diastole, indicating high resistance or obstruction.
Diffraction	Bending of waves around obstacles.
Digital-to-Analog (D-to-A) Converter	Converts digital signals to analog for display or transmission.
Directional Power Doppler	Power Doppler showing flow direction.
Divergence	Spreading of the ultrasound beam in the far field.
Doppler Effect	Change in frequency due to relative motion between source and observer.
Doppler Equation	Formula for frequency shift: f = (2 * v * f * cos) / c, where v=velocity, f=frequency, =angle, c=speed.
Doppler Ghosting	Artifact from vessel wall motion mimicking flow.
Doppler Phantom	Test object simulating blood flow for Doppler calibration.
Duplex	Combination of B-mode imaging and Doppler.

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Term	Definition
Duty Factor	Fraction of time the transducer is transmitting.
Dynamic Range	Ratio of largest to smallest signal the system can handle.
Edge Shadowing	Shadowing from curved surfaces due to refraction.
Elevational Resolution	Resolution in the slice thickness plane.
Energy Gradient	Difference in energy driving blood flow (pressure, gravity, etc.).
Enhancement	Increased echogenicity distal to fluid-filled structures due to low attenuation.
Ensemble Length	Number of pulses per color line; affects velocity accuracy.
Envelope (Enveloping)	Tracing the outline of the spectral Doppler waveform.
Far Zone	Region beyond focal zone where beam diverges (Fraunhofer zone).
Fast Fourier Transform (FFT)	Algorithm to analyze frequency content in Doppler signals.
Fill-in Interpolation	Filling gaps in color Doppler data for smoother display.
Flash Artifact	Color artifact from motion like breathing or probe movement.
Flow	Volume of blood moving per unit time.
Focal Zone	Area where beam is narrowest and resolution best.
Footprint	Contact area of transducer with skin.
Frame Rate	Number of images per second; affects temporal resolution.
Frequency	Number of cycles per second (Hz); higher improves resolution but increases attenuation.
Frequency Compounding	Using multiple frequencies to reduce speckle.
Frequency Shift	Difference in transmitted and received frequency due to Doppler.
Fresnel Zone	Near field where beam is complex.
Friction	Resistance to flow at vessel walls.
Fundamental Frequency	Primary transmitted frequency.
Grating Lobes	Unwanted side beams from array transducers.
Gravitational Potential Energy	Energy due to position in gravity field affecting venous flow.
Grayscale Test Objects	Phantoms for calibrating B-mode imaging.
Harmonics	Multiples of fundamental frequency; used in tissue harmonic imaging.
Hemodynamics	Study of blood flow dynamics.
Horizontal Calibration	Testing lateral distance accuracy.
Hue	Color attribute in color Doppler representing direction or velocity.
Huygens Principle	Each point on a wavefront acts as a source of secondary waves.

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Term	Definition
Hydrophone	Device to measure ultrasound pressure fields.
Hyperechoic	Brighter than surrounding tissue.
Hypoechoic	Darker than surrounding tissue.
Impedance	See Acoustic Impedance.
Inertia	Resistance of fluid to change in motion.
Innervated	Supplied with nerves.
In-Phase	Waves aligned in time.
Intensity	Power per unit area of ultrasound beam.
Interface	Boundary between two media with different impedances.
Intensity Reflection Coefficient	Fraction of intensity reflected at an interface.
Intensity Transmission	Fraction of intensity transmitted through an interface.
Inversely Related	When one variable increases, the other decreases.
Kinetic Energy	Energy of motion in blood flow.
Lab Accreditation	Certification of vascular labs (e.g., IAC).
Laminar Flow	Smooth, layered flow in normal vessels.
Lateral Resolution	Ability to distinguish structures perpendicular to the beam.
Law of Conservation of Energy	Energy is neither created nor destroyed; applies to blood flow.
Lead Zirconate Titanate (PZT)	Common piezoelectric material.
Linear Phased Array	Transducer for electronic steering and focusing.
Linear Sequenced Array	Transducer firing groups of elements sequentially.
Liquid Crystal Display (LCD)	Modern display for ultrasound machines.
Longitudinal Waves	Waves where particle motion is parallel to propagation.
Luminance	Brightness in color display.
M-Mode	Motion mode; displays depth vs. time for moving structures.
Master Synchronizer	Controls timing in ultrasound system.
Matching Layer	Layer on transducer to improve energy transfer to tissue.
Mechanical Index	Indicator of cavitation potential.
Mechanical Scanhead	Transducer with moving parts for scanning.
Medium	Substance through which sound travels.
Mirror Image	Artifact where structure appears on both sides of a reflector.
mmHg	Millimeters of mercury; unit for blood pressure.
Multipath	Echoes taking indirect paths, causing artifacts.
Nyquist Limit	Half the PRF; maximum detectable velocity without aliasing.

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Term	Definition
Peak Systolic Velocity (PSV)	Highest velocity during systole; used to grade stenosis.
Perfusion	Blood flow to tissues.
Phasicity	Variation in flow with respiration or cardiac cycle.
Plethysmography	Measurement of volume changes; used in vascular testing.
Poiseuille's Law	Law describing the flow of a fluid through a cylindrical tube, stating that flow rate is proportional to the pressure difference and the fourth power of the radius, and inversely proportional to the length and viscosity; key in vascular resistance calculations.
Power Doppler	Displays flow amplitude rather than velocity; sensitive to slow flow.
Pressure Gradient	Difference in pressure driving flow.
Pulsatility Index (PI)	(PSV - EDV) / mean velocity; measures pulsatility.
Pulsed Wave (PW) Doppler	Doppler with range gating for depth-specific flow.
Resistance	Opposition to blood flow; high in arterioles.
Resistive Index (RI)	(PSV - EDV) / PSV; indicates downstream resistance.
Retrograde	Flow in the reverse direction.
Reynolds Number	Dimensionless number indicating the likelihood of turbulent flow, calculated as (density * velocity * diameter) / viscosity; critical in assessing vascular flow patterns.
Ring Down	Artifact from gas or metal causing continuous echoes.
Slice Thickness Artifact	Out-of-plane structures appearing in image.
Spectral Broadening	Widening of Doppler spectrum indicating turbulence.
Stenosis	Narrowing of a vessel lumen.
Thrombosis	Formation of a blood clot in a vessel.
Triphasic Waveform	Normal arterial waveform with forward, reverse, forward components.
Turbulent Flow	Chaotic flow patterns, often post-stenosis.
Valsalva Maneuver	Bearing down to increase intrathoracic pressure; tests venous competence.
Velocity	Speed and direction of blood flow.
Venous Insufficiency	Inadequate venous return due to valve failure.
Viscosity	Internal friction of blood; affects flow resistance.
Wall Filter	Removes low-frequency clutter from Doppler signals.
Ankle-Brachial Index (ABI)	Ratio of the ankle systolic pressure to the brachial systolic pressure, used to diagnose peripheral artery disease; normal range is 1.0-1.4, with <0.9 indicating disease.

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Term	**Definition**
Vascular Resistance	Opposition to blood flow in vessels, influenced by vessel diameter, length, and blood viscosity; measured indirectly via Doppler indices.
Pulse Volume Recording (PVR)	Non-invasive test measuring volume changes in limbs to assess arterial flow; often used with ABI.
Segmental Pressures	Blood pressure measurements taken at different levels of the limb to localize arterial disease.
Waveform Analysis	Evaluation of Doppler waveforms (e.g., triphasic, biphasic) to assess vascular health.